

CTTB NETWORK INFRASTRUCTURE

City of Ten Thousand Buddhas | Core Services & Physical Topology
LAN 10.11.0.0/16 | DNS .cttb | Proxy srv-gw:8080 | 2026-04-29

LAYER 1 | WAN UPLINKS

INTERNET

com7 (172.30.7.2/24)

ISP #1 wt:10

com20 (172.30.20.2/24)

ISP #2 wt:30

com10 (172.30.10.2/24)

ISP #3 wt:30

LAYER 2 | NETWORK EDGE

sw-mpoe | 10.11.12.30 | Main Point of Entry

srv-gw

10.11.1.1 | GATEWAY / FIREWALL

FIREHOL

Stateful firewall + timed Internet

SQUID

Forward proxy & cache on :8080

E2GUARDIAN

Content filter with SSL MITM

MULTI-WAN LB

Round-robin 10/30/30 weighted

NTP

time.cttb for all clients

LAYER 3 | SERVER ROOM

sw-ao-srvrm | 10.11.12.31 | Server Room Aggregation Switch

srv-vm

10.11.1.3 | VIRTUALIZATION HOST

- ▶ 16 LXD containers on ZFS
- ▶ bond0: 802.3ad LACP 4xGbE → lxdbr0
- ▶ ZFS raidz2: 4x Samsung 850 EVO 1TB
- ▶ NUT netclient (UPS slave → srv-nas)

srv-nas

10.11.1.5 | NETWORK STORAGE

- ▶ 7 LXD containers on ZFS
- ▶ bond0: 802.3ad LACP 4xGbE → lxdbr0
- ▶ ZFS raidz2: 6xSAS + NVMe SLOG
- ▶ NUT master-slave: IUT master: Eaton 5S 1500VA UPS

↔ NUT master-slave

Datasets: lxd, logs(1T), kvm, nethomes(1T)

STANDBY REPLICAS

Currently offline

- ▶ srv-bk-gw 10.11.1.9 → Gateway replica
- ▶ srv-bk-vm 10.11.1.7 → VM host replica (ZFS raidz2)
- ▶ srv-bk-nas 10.11.1.11 → NAS replica

LAYER 4 | CORE SERVICES (LXD containers, bridged via lxdbr0)

srv-vm CONTAINERS (10.11.1.3)

AUTHENTICATION

- lxc-ldap .1.25 OpenLDAP dc=cttb

MONITORING

- lxc-mon .1.26 System monitoring

DNS

- lxc-ub-adult .1.29 Unbound (adult)
- lxc-ub-igdvs .1.28 Unbound (schools)
- lxc-dnsmasq .1.19 DNSmasq cache

PRINTING

- lxc-cups-cttb .1.36 Campus-wide
- lxc-cups-dvbs .1.37 Boys School
- lxc-cups-dvgs .1.38 Girls School

VOIP

- lxc-asterisk .1.32 PBX + TFTP phones

COLLABORATION

- lxc-wiki .1.31 Internal wiki
- lxc-blogger .1.42 Blog platform

SSH ACCESS

- lxc-jumpbox .1.33 Bastion / admin gw

APPLICATIONS

- lxc-sltp .1.39 Sanskrit platform
- lxc-sltp-git .1.40 SLTP Git backend
- lxc-drbu-sis .1.41 DRBU Student Info

srv-nas CONTAINERS (10.11.1.5)

FILE STORAGE

- lxc-fs .1.18 NFS /nethomes (1TB zvol, privileged)

CENTRALIZED LOGGING

- lxc-log .1.20 Syslog (bind-mount /data/logs, 1TB)

SOURCE CONTROL

- lxc-git .1.21 Git server

DEPLOYMENT PIPELINE

- lxc-debmirror .1.22 Apt mirror (Ubuntu, Koha, VBox)
- lxc-pxe .1.23 PXE: TFTP + Apache preseed

METRICS

- lxc-metrics .1.24 Metrics collection

LIBRARY SYSTEM

- lxc-koha .1.27 Koha ILS (library.igdvs.cttb)

CLIENT LIFECYCLE

① PXE ROM

Client powers on

② DHCP

IP + next-server

③ lxc-pxe

TFTP lpxellinux.0

④ Boot Menu

default / raid1/6

⑤ Preseed

apt.cttb autoinstall

⑥ READY

Client joins LAN

▼ Running client connects to these services:

DNS

.1.29/28 Unbound

AUTH

.1.25 LDAP login

HOME

.1.18 NFS mount

PRINT

.1.36-38 CUPS

WEB

gw:8080 Squid+E2G

SYSLOG

.1.20 Centralized

TIME

.1.1 NTP

LAYER 5 | NETWORK ZONES (subnets within 10.11.0.0/16)

10.11.1.x INFRASTRUCTURE | Servers & containers | Always on | No filter

.9.x DVGS

Girls School | filter:75 MITM | 7-22 daily

.10.x DVBS

Boys School | filter:75 MITM | Time-windowed

.8.x ADMIN

Restricted | filter:50 | Always on

.15.x DRBU

University | filter:400 | Always on

.12.x SWITCHES | 80+ managed HP/Cisco

.14.x PHONES | Student phones f:400

.19.x LAPTOPS | Student f:75

.21-25 STAFF | Faculty/home f:400

CONTENT FILTERING (E2Guardian on srv-gw)

GROUP	SUBNET	LIMIT	BYPASS	APPLIES TO	TIMED INTERNET SCHEDULES
Adult	default	400	300	Staff, DRBU, faculty	DVBS School 6-10:30, 11:30-16:30 M-F
Adult No Bypass		175	0	15 specific staff IPs	DVBS Comm Ctr 12-14:30, 16-17, 18-22 M-F
Restricted	10.11.8.x	50	0	Admin subnet	DVGS 7:00-22:00 daily
IGDVS	10.11.9-10.x	75	MITM	Schools + SSL interception	DVGS CS Lab 8-10:30, 12-18:30 M-F

LAYER 6 | CAMPUS SWITCH FABRIC (10.11.12.x | 80+ managed switches)

sw-mpoe .12.30 → sw-ao-srvrm .12.31

WAN & server room uplinks

DVGS

Girls School

CS Lab .12.44
Dorm .12.148
Library .12.97
ES Admin .12.206
+1 more

DVBS

Boys School

CS Lab .12.179
Main Ofc .12.209
Comm Ctr .12.158
Juan Ofc .12.37
+1 more

1234

Admin Bldg

Machine Rm .12.172
Net Room .12.38
Spike Cube .12.216

AO

Admin Office

Server Rm .12.31
Attic .12.167
Front Desk .12.156

ToB

Tower of Bliss

Center .12.34
Elevator .12.165
Financial .12.241
EnvSci .12.285

DRBU

University

CS Lab .12.242
Lib Up .12.93
Lib Rm .12.141
Comm .12.169
+1 more

OTHER

Campus-wide

JGH .12.131
Laundry .12.211
Restaurant .12.91
Solar .12.199

DNS ZONE RECORDS (.cttb domain)

RECORD	IP	SERVICE	RECORD	IP	SERVICE
gw.cttb	10.11.1.1	Gateway/FW/NTP	mon.cttb	10.11.1.26	Monitoring
srv-vm.cttb	10.11.1.3	VM host	library.igdvs.cttb	10.11.1.27	Koha library
srv-nas.cttb	10.11.1.5	NAS	ub-igdvs.cttb	10.11.1.28	Unbound (schools)
fileserver.cttb	10.11.1.18	NFS home dirs	ub-adult.cttb	10.11.1.29	Unbound (adult)
dnsmasq.cttb	10.11.1.19	DNS cache	wiki.cttb	10.11.1.31	Wiki
log-srv.cttb	10.11.1.20	Syslog	asterisk.cttb	10.11.1.32	VoIP PBX
git.cttb	10.11.1.21	Git server	cups-cttb.cttb	10.11.1.36	Print server
apt.cttb	10.11.1.22	Apt mirror	lxc-sltp.cttb	10.11.1.39	Sanskrit
pxe.cttb	10.11.1.23	PXE / TFTP	sis.drbu.cttb	10.11.1.41	Student info
ACCESS: ldap.cttb	Jumpbox: ssh administrator@cttb (password) Infrastructure: pubkey only Containers: lxc exec <name> bash time.cttb		4mlt0f0 (sudo)	10.11.1.1	NTP (alias)

WHAT RUNS WHERE?

NEED TO...	GO TO	IP
Debug internet/firewall	srv-gw	10.11.1.1
Manage LXD containers	srv-vm/srv-nas	.1.3/.1.5
Reset user password	lxc-ldap	.1.25
Fix DNS	lxc-ub-adult/igdvs	.1.29/.28
Fix printing	lxc-cups-*	.1.36-38
PXE boot machine	lxc-pxe	.1.23
NFS home dirs	lxc-fs	.1.18
Centralized logs	lxc-log	.1.20
Update apt mirrors	lxc-debmirror	.1.22
VoIP / phones	lxc-asterisk	.1.32
Library (Koha)	lxc-koha	.1.27
Monitoring	lxc-mon/metrics	.1.26/.24
Git repos	lxc-git	.1.21
SSH from outside	lxc-jumpbox	.1.33